# VetLink PWA — Cloud Deployment & Configuration Guide

Actionable Setup Guide for Rus (Repository & Cloud Infrastructure Lead) | EARIST Capstone 2026

## 1. Overview & System Architecture

VetLink PWA operates with a decoupled architecture: the frontend is hosted on **Vercel** (vetlink-pwa.vercel.app), while the AI inference engine and centralized MySQL database connect through a **Render** cloud web service. To allow all clinic devices (phones, laptops) to access centralized AI inference and database sync without relying on a local machine, the cloud backend URL must be injected into the Vercel production build.

## 2. Step-by-Step Configuration Instructions for Rus

| Step | Action Required | Technical Notes |
|---|---|---|
| **1. Retrieve Render Backend URL** | Log in to **dashboard.render.com** > Select the VetLink web service > Copy the public URL (e.g. https://vetlink-backend.onrender.com). | Ensure status is **Live (Green)**. Do not include a trailing slash /. |
| **2. Add Variable in Vercel** | Log in to **vercel.com** > Project **vetlink-pwa** > **Settings** > **Environment Variables**.<br>Add Key: VITE_API_BASE_URL<br>Value: https://vetlink-backend.onrender.com | Check all 3 environments: **Production**, **Preview**, and **Development**. |
| **3. Trigger Redeploy on Vercel** | In Vercel, navigate to **Deployments** > Click the ... menu next to the latest commit > Click **Redeploy** (or merge the upcoming pull request). | Vite bakes import.meta.env.VITE_API_BASE_URL during build time; a redeploy is mandatory. |
| **4. Verify Live Connectivity** | Open **https://vetlink-pwa.vercel.app** > Open DevTools (F12) > Network tab > Perform a test scan. Verify requests hit the Render backend with HTTP 200. | FastAPI CORS is already configured to accept * origins. |

## 3. Free-Tier Behavior & Clinical Testing Best Practice

VetLink runs smoothly on Render Free Tier. However, Free Tier instances **spin down (sleep)** after 15 minutes of inactivity. The initial request takes approximately **30 to 50 seconds** to wake the container. Subsequent requests execute at full speed (0.04s inference). **Recommendation:** 5 minutes before clinic testing or presentation, open the backend URL in a browser once to wake the server.

## 4. Security & Environment Variable Safety

• **Git Protection:** The .gitignore file strictly ignores .env and .env.*. Never hardcode live database credentials or session secrets in source code.
• **Template Reference:** Developers should refer to .env.example for standard variable keys.
• **Authorization on Vercel:** Only team members with deploy permissions (Rus) can authorize production deployments for the VetLink team.